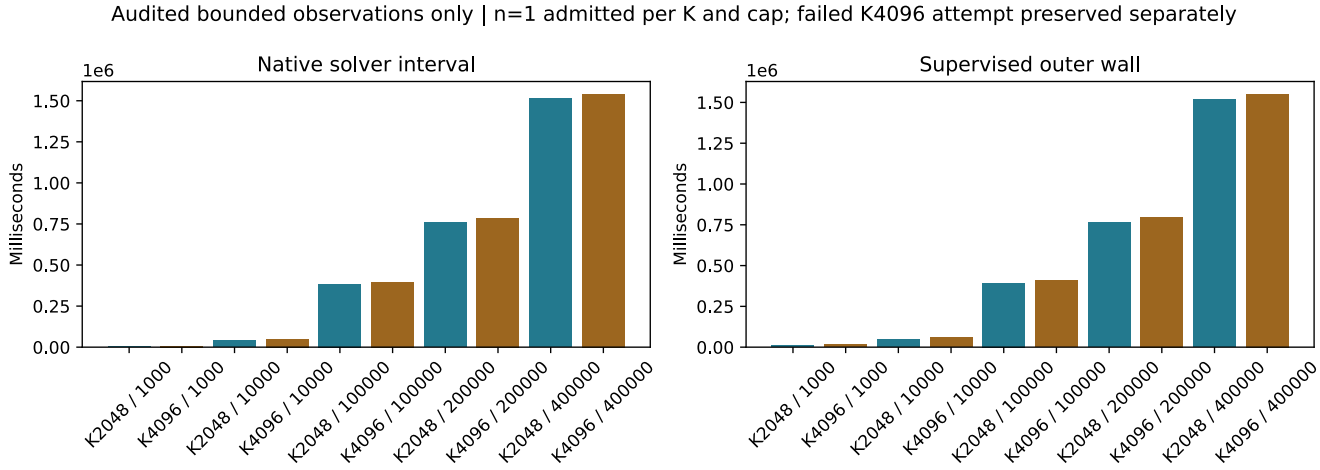
Native N83 factor-base continuation: 400000-conflict caps

Public synthetic a=0 icv1-f2m83-tm6151469093347-debefd74 over x^83+x^45+x^2+x+1; subgroup order24178516392307962166856 89. The positive-trace a=1 N83 cold holdout is distinct. Source-linked protocol, raw records and S3 verification are linked in REPORT.md beside this PDF.

The user requested another higher cap for K2048/K4096 and all raw evidence in setec-astronomy-artifacts. This separately committed protocol raises200000→400000conflicts per base, with2400seconds per worker and2490seconds per mechanical command. K2048 runs once, is assessed, then K4096 runs independently. After an assessed inspection failure, a separately frozen K4096 repeat uses3000seconds per mechanical command; maximum cumulative worker allocation7200seconds covers all3attempts. Readiness/build/test/ replay/render/storage are separate overhead. Historical1000/10000/100000/200000 observations are immutable and not pooled as repetitions.

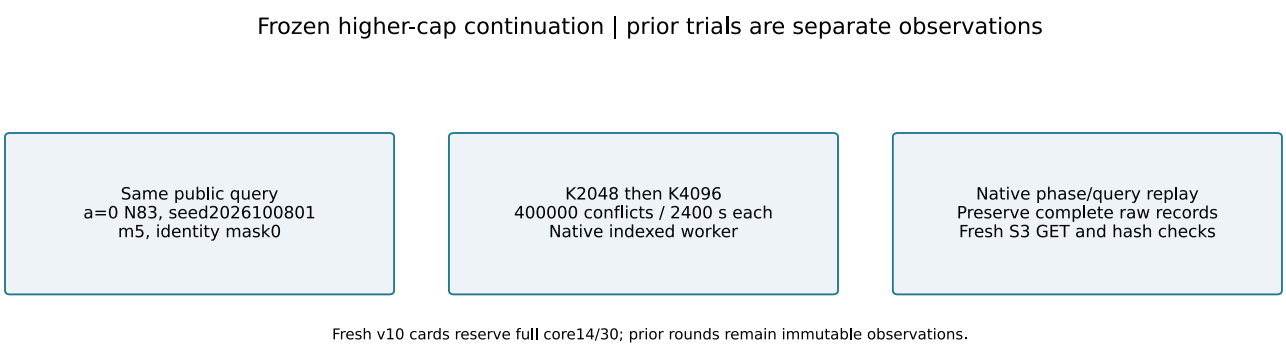


Native Rust derives solver interval as return.elapsed_ms-enter.elapsed_ms. Outer wall includes supervision/ setup and is separately recorded. A killed prefix has no complete terminal counter or solver interval. Descriptive n=1 admitted per K/cap, no A/A interval, admitted timing/base ranking, fastest-tool claim or IC/rho ratio.

Recorded new outcomes (native audit; n=1 each): K2048: PASS_NATIVE_BOUNDED_ZERO_RELATION_REPLAY; terminal conflicts 400000, models 0, relations 0, rank 0; solver 1513432.75291 ms, outer 1519074.0337 ms. K4096: PASS_NATIVE_BOUNDED_ZERO_RELATION_REPLAY; terminal conflicts 400000, models 0, relations 0, rank 0; solver 1540183.375695 ms, outer 1550621.509279 ms.

Recorded outcomes, validation and custody

K / cap	Conflicts	Native audit	Solver ms
K2048 / 1000	1000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	4861.868809
K4096 / 1000	1000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	7814.674415
K2048 / 10000	10000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	42413.639201
K4096 / 10000	10000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	50433.062636
K2048 / 100000	100000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	383937.390858
K4096 / 100000	100000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	395987.838978
K2048 / 200000	200000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	759947.847210
K4096 / 200000	200000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	786700.645545
K2048 / 400000	400000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	1513432.752910
K4096 / 400000	400000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	1540183.375695



Retained worker SHA2563a31bd2bedee5519d30b517f39ec27f72561ce898213657688c55b51bbcdf7b6 is not rebuilt. Its9watched solver/encoding/example/adapter files match660f0f4b10. New source d216680177 identifies guard/protocol, not a rebuilt worker. Same retained single public query, public_x_hash seed2026100801, m5, identitymask0 and exact public K2048/K4096 inputs.

Immutable guard-v10 SHA25676155c48c44e0af5ccb06708f1399fc4e3aa1c6ceb51b72bcccaeeb7b6ca2b11 only extends natural admission to400000conflicts/2400s. Prior200000/1200 remains accepted; zero and just-above-limit reject. Witness100000/900 and external1000000/900 stay unchanged.18unit+7card tests pass before timing. CPU reservation and corrected native summary classification are unchanged.

Prior library4398pass/118ignored validation is inherited. Post-base-merge full serial had4398pass/9socket-denied/118ignored, followed by elevated38-test affected module pass covering all9denials, plus19exporter/2model checks. Failed and successful logs remain archived. No newly clean single full-suite run is claimed; no library/solver change here.

Same immutable Docker image541096ae9b05…,1CPU quota/4GiB/noextra swap, networknone, readonly root/source/input/worker, capdropALL/no-new-privileges/64PIDs. Full core14/30 reserved, runCPU14; unchanged strict topology/cgroup/SMT checks, exclusive lock, movable-thread eviction/restoration, quiet thresholds and bounded readiness wait. Actual before/after conditions and cleanup remain raw evidence. No peer fixed cpuset is changed or job stopped. Fixed kernel threads and missing worker context-switch/fault counters preclude automatic L2. No builds/tests/heavy work from this task during measured arms.

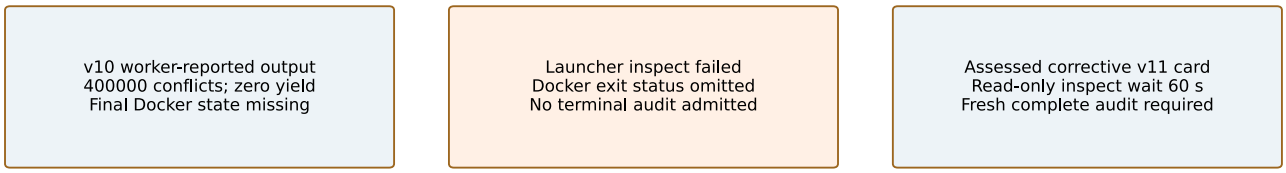
Director compiles/assesses; mechanical executes exact frozen cards with actual stdout under noclobber, fresh write-once paths. Every refusal/failure is assessed before a new plan; no automatic timed retry. Docker timeouts can leave delayed Created containers; later reconciliation is retained separately without rewriting the original snapshot. Prior setup failures remain at their upstream content addresses; every new attempt is in this raw archive.

Native audit checks original-field/group queryaG+bQ, phase order, input/source/ELF/stream custody, budgets, exit/OOM and cleanup. Positive natural models remain fail-closed pending independent original-group/rank/scalar checks. Bounded unknown is not UNSAT or absent decomposition. No useful natural yield, best base, scalar recovery or cold IC/rho winner is inferred from a capped zero result.

CryptoMiniSat5.17.0 remains faster on the separate verified witness corpus, with distinct CPU/wall timer scopes retained upstream. No fresh matched external comparison for these capped unknown natural queries; no fastest claim. Canonical cold/rho/identity graphs stay unchanged absent their separate evidence gates. Same-source SVG/150dpiPNG, inspected PDF, benchmark copies and complete raw S3 evidence are included. Research, replay and derived values are native; Python only lays out recorded data and inspects PDF.

Preserved orchestration failure and corrective repeat

K4096 orchestration failure and separately frozen repeat



Original failure and cost retained; no pooling or replacement of its records.

K4096-v10 reported400000conflicts/zero yield but failed its15-second Docker inspection RPC: final exit/OOM/outer/ audit missing. Command1598.372932784s and raw worker-reported1549664.280075ms remain archived, not admitted as a complete-run clock. After assessment, one corrective K4096 repeat is separately frozen atcad91042bf using guard-v11 SHA2564c3d3f6658134d68da132b2efb0948195ef6b6031a40168dc875e25d946f765b. Inspect wait alone extends15→60s; no RPC retries.19unit+7card tests pass. Same worker/query/resources,2400s worker limit,3000s mechanical ceiling; maximum cumulative worker allocation7200s includes all3attempts. Table has n=1 admitted per K/cap; K4096 has two total attempts including its preserved orchestration failure. Original v10 records remain intact; later absence was separately confirmed.

Terminology correction: v10 omitted Docker’s exit status. Timeout is inferred from stalled RPCs and the15-second policy, not a preserved exit124 observation. V11′s bounded wait and better errors are mitigation, not a universal root-cause proof. Original frozen records remain unchanged.